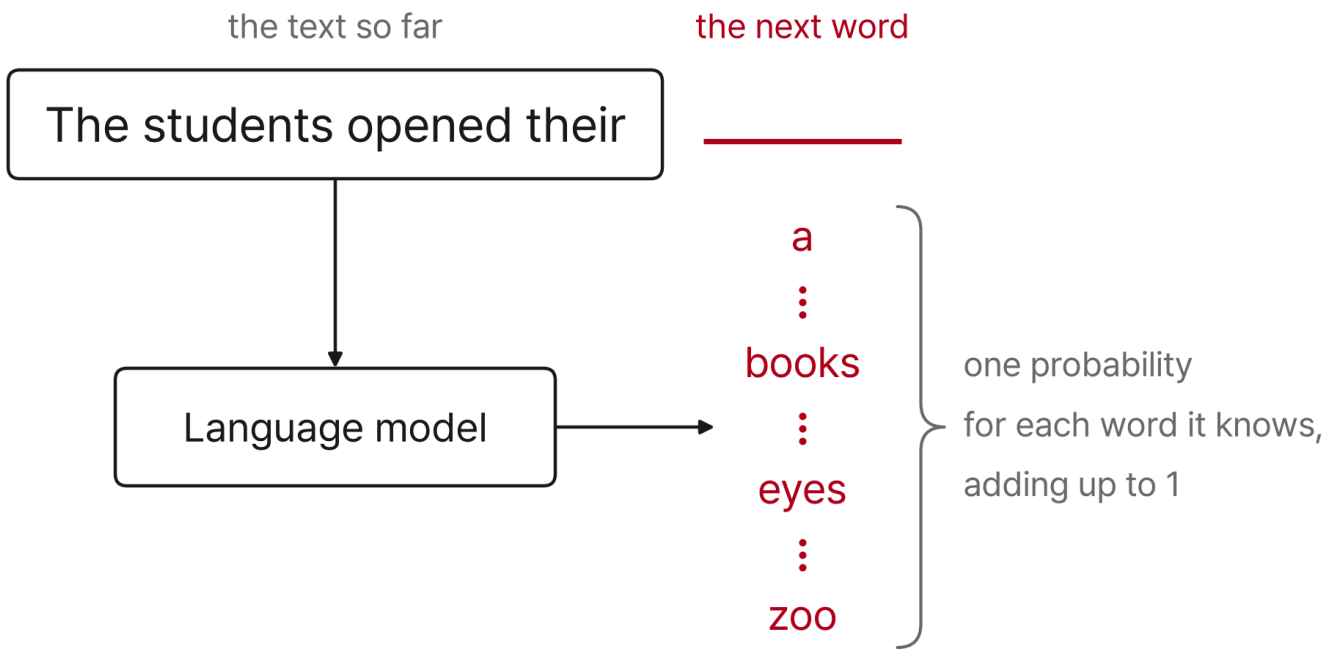
Data & AI

14. Large Language Models

Dasaem Jeong · Art & Technology, Sogang University

Last Time



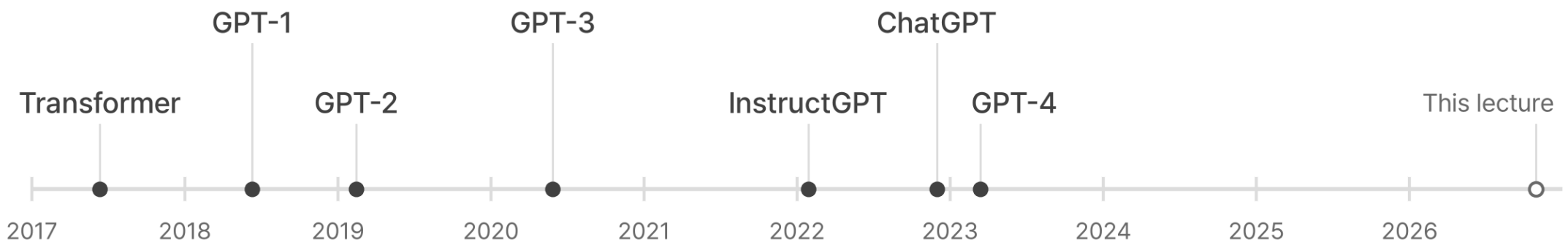
- To fill the seven blanks: facts, grammar, who is who, the writer's opinion, arithmetic
- Larger models gave the expected word a higher probability in most blanks
- The names RNN: 23,451 parameters, trained on 28,829 names

Today

- GPT models from 2018 to 2023: what each one wrote
- What changed in each model, and why its answers got better

GPT

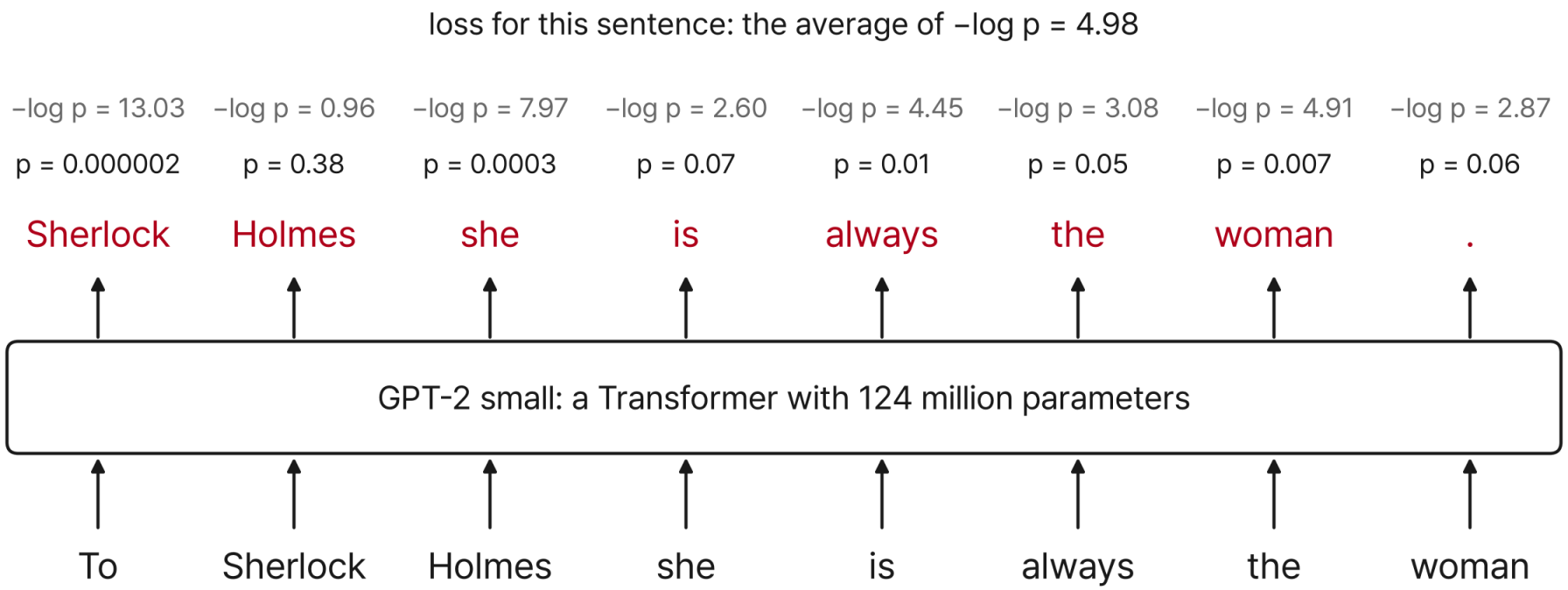
Generative Pre-trained Transformer



- OpenAI's GPT models from 2018 to 2023, after the Transformer (2017)
- The same task in every one: predict the next word

The Block in the Middle

GPT-2 small on the first sentence of "A Scandal in Bohemia"



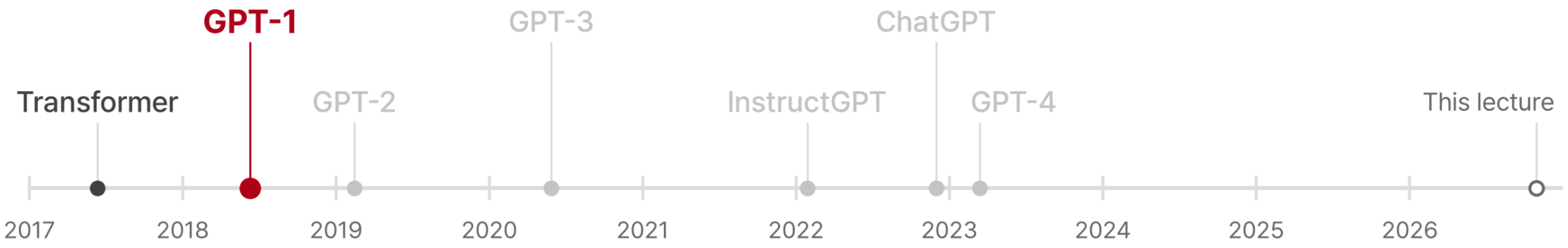
- The names RNN: an RNN in the middle, reading one letter at a time
- GPT: a Transformer in the middle, an architecture Google published in 2017 for translation

Vaswani et al. (2017), Attention Is All You Need, NeurIPS 2017 · figure: our run of GPT-2 small

GPT-1

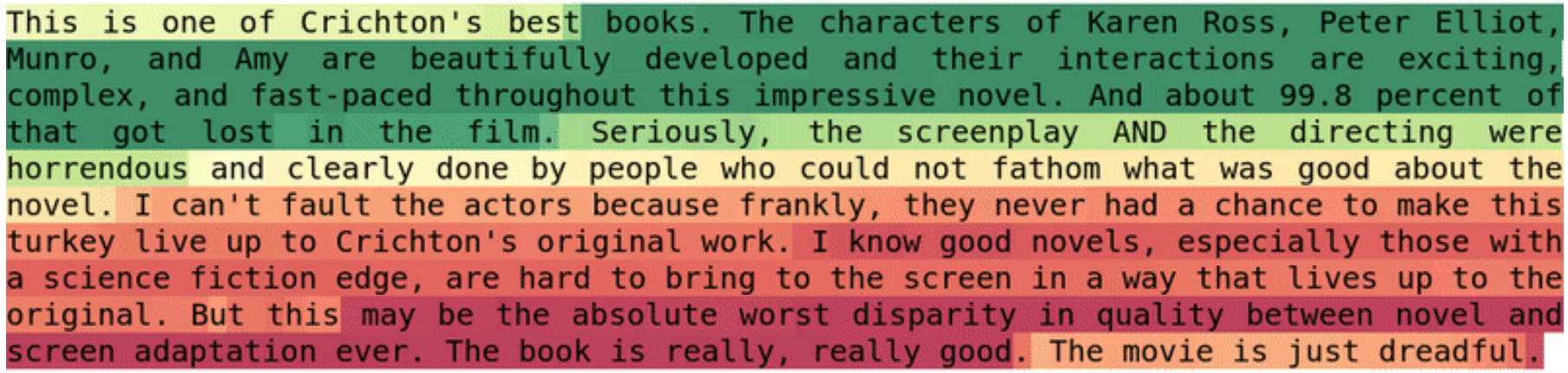
June 2018 · 117 million parameters · 7,000 unpublished books

Radford, Narasimhan, Salimans, Sutskever (2018), Improving Language Understanding by Generative Pre-Training



Sentiment Neuron

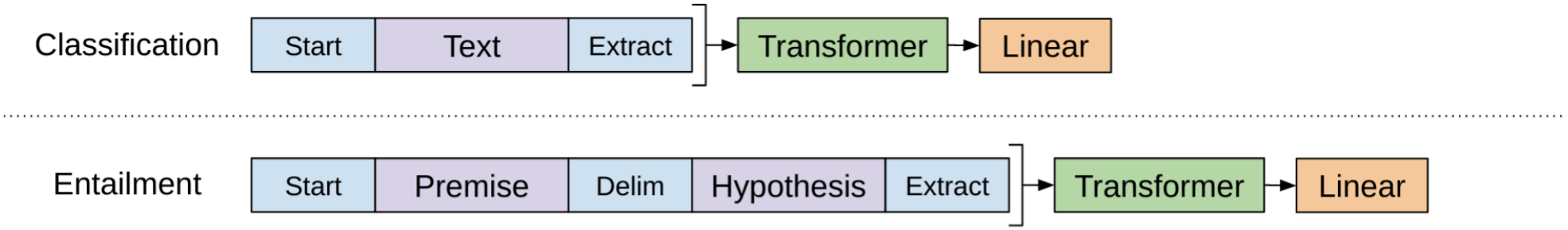
Radford et al. (2017), Learning to Generate Reviews and Discovering Sentiment



- A network trained only to predict the next character of 82 million Amazon reviews
- Color: the "sentiment neuron", one of its 4,096 hidden values (green high, red low)
- GPT-1, a year later, was developed following this work (OpenAI blog, 2018)

Pre-training and Fine-tuning

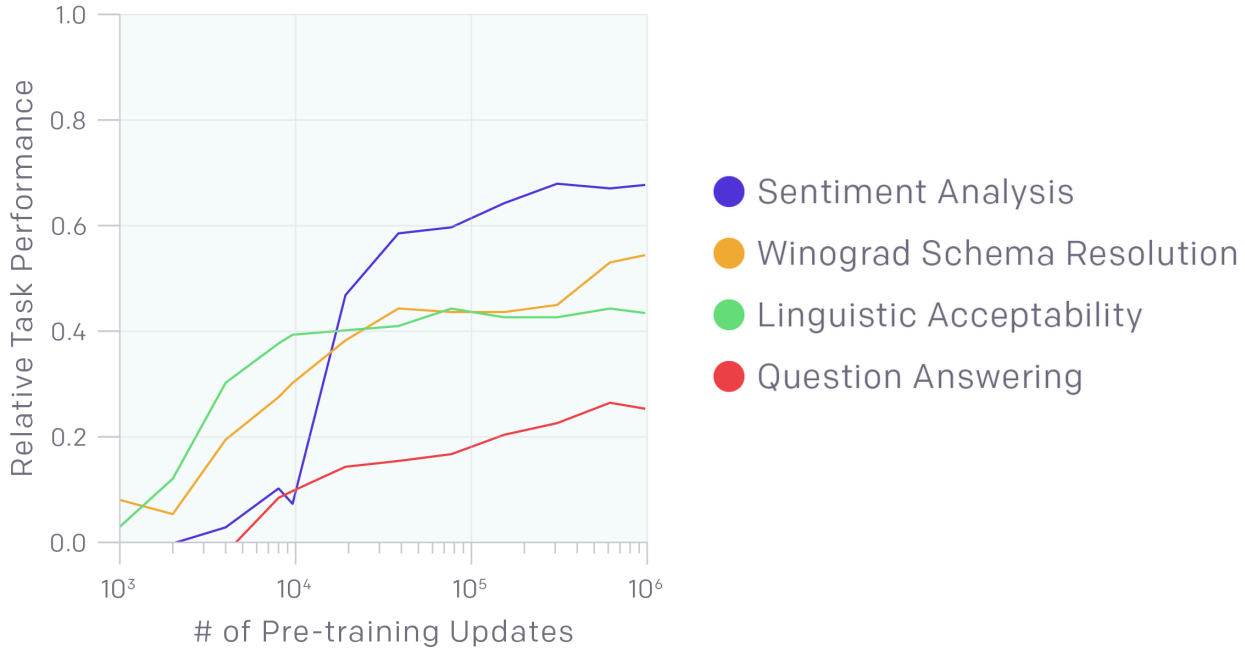
Radford et al. (2018), Figure 1: one pre-trained Transformer, a new Linear layer for each task



- Pre-training: predict the next word in 7,000 books, for a month on 8 GPUs
- Fine-tuning: a new last layer, trained on one labeled task
- Assignments 1 and 2: ResNet18 trained on ImageNet, a new last layer for your photos
- Result: the best score on 9 of 12 tasks; 14.8 points lower on average without pre-training

Zero-Shot Behaviors

Radford et al. (2018), Section 5: four tasks with no fine-tuning and no examples



- Sentiment: compare the probabilities of "positive" and "negative" after a review and "very"
- Left to right: longer pre-training · 0: random guessing, 1: the best 2018 system
- The paper's guess: the model learns these tasks to predict the next word better

Radford et al. (2018), Section 5 and Figure 2 · figure: OpenAI blog (2018), legend moved to the right and enlarged

Text Written by GPT-1

Our run: the prompt OpenAI later used for GPT-2, first try (seed 0), top-k 40

Prompt, written by a person

> In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

GPT-1, first try

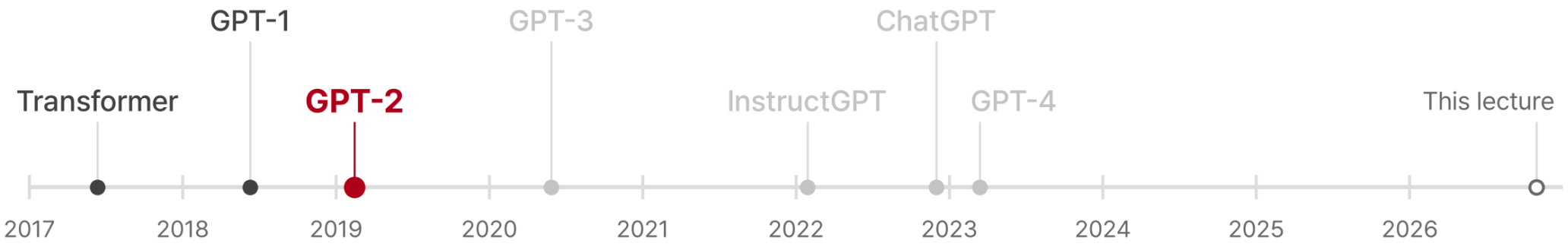
> most of the scientists who came to study the animals had been experimenting with other languages . the humans did n't , of course , but had come up with a hypothesis that the human species had originated from the andes and were now living in an alternate universe .
> it occurred to one of the scientists that the concept of ' parallel universes ' had never been thought to be a possibility among humans . scientists began to think that the two parallel universes were a combination of the scientific method and the other , [...]

Model: openai-community/openai-gpt, run by the course staff

GPT-2

February 2019 · 1.5 billion parameters · 40 GB of web text

Radford, Wu, Child, Luan, Amodei, Sutskever (2019), Language Models are Unsupervised Multitask Learners



A Sample from GPT-2

OpenAI blog, Better Language Models and Their Implications, 14 February 2019

Prompt, written by a person

> In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

GPT-2, the best of 10 tries

> The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.
> Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.
> Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. [...]
> Dr. Pérez believes that the unicorns may have originated in Argentina, where the animals were believed to be descendants of a lost race of people who lived there before the arrival of humans in those parts of South America. [...]

A Sample from GPT-2

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Prompt, written by a person

> In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

GPT-2, the best of 10 tries

> The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.
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> Dr. Pérez believes that the unicorns may have originated in Argentina, where the animals were believed to be descendants of a lost race of people who lived there before the arrival of humans in those parts of South America. [...]

The Same Prompt in GPT-2 XL

Our run: the same size as the model in the blog post (1.5B), first try (seed 0)

GPT-2 XL, first try, not chosen

> The animals, known as Andean unicorns, lived during the Cenozoic Era, approximately 120 million years ago.
> During the Cenozoic Era, a portion of the Andes Mountain range in South America was covered by a thick blanket of lush forest, with no signs of human human encroachment. This particular valley, which was previously a mountain range, was also thought to be devoid of other life.
> The researchers used data from the ancient DNA project to identify a "genome" of the unicorns, which revealed that the unicorns have a close genetic relationship to other domestic horses. The research, led by paleontologist Sergio Caballero, was reported in Biology Letters.

Model: openai-community/gpt2-xl, top-k 40, temperature 1.0, run by the course staff

The Release of GPT-2

OpenAI, February to November 2019



AI (artificial intelligence)

This article is more than **7 years old**

New AI fake text generator may be too dangerous to release, say creators

The Elon Musk-backed nonprofit company OpenAI declines to release research publicly for fear of misuse

Alex Hern

Thu 14 Feb 2019 17.00 GMT

> "Due to our concerns about malicious applications of the technology, we are not releasing the trained model." (OpenAI, 14 February 2019)

- Released in stages: 124M in February, 355M in May, 774M in August, 1.5B in November

Question Answering Without Training

Radford et al. (2019), Table 5: the 13 answers to which GPT-2 gave the highest probability

Question	Generated Answer	Correct	Probability
Who wrote the book the origin of species?	Charles Darwin	✓	83.4%
Who is the founder of the ubuntu project?	Mark Shuttleworth	✓	82.0%
Who is the quarterback for the green bay packers?	Aaron Rodgers	✓	81.1%
Panda is a national animal of which country?	China	✓	76.8%
Who came up with the theory of relativity?	Albert Einstein	✓	76.4%
When was the first star wars film released?	1977	✓	71.4%
What is the most common blood type in sweden?	A	✗	70.6%
Who is regarded as the founder of psychoanalysis?	Sigmund Freud	✓	69.3%
Who took the first steps on the moon in 1969?	Neil Armstrong	✓	66.8%
Who is the largest supermarket chain in the uk?	Tesco	✓	65.3%
What is the meaning of shalom in english?	peace	✓	64.0%
Who was the author of the art of war?	Sun Tzu	✓	59.6%
Largest state in the us by land mass?	California	✗	59.2%

- Never trained on question answering; example questions and answers in the prompt
- 4.1% correct, the smallest GPT-2 at most 1%; question-answering systems then: 30 to 50%

Radford et al. (2019), Table 5 and Section 3.8

Tasks Inside Web Text

Radford et al. (2019), Table 1: translations found in the text GPT-2 learned from

”I’m not the cleverest man in the world, but like they say in French: **Je ne suis pas un imbecile [I’m not a fool].**

In a now-deleted post from Aug. 16, Soheil Eid, Tory candidate in the riding of Joliette, wrote in French: ”**Mentez mentez, il en restera toujours quelque chose**,” which translates as, ”**Lie lie and something will always remain.**”

“I hate the word ‘**perfume**,”’ Burr says. ‘It’s somewhat better in French: ‘**parfum**.’

If listened carefully at 29:55, a conversation can be heard between two guys in French: “**-Comment on fait pour aller de l’autre coté? -Quel autre coté?**”, which means “**- How do you get to the other side? - What side?**”.

If this sounds like a bit of a stretch, consider this question in French: **As-tu aller au cinéma?**, or **Did you go to the movies?**, which literally translates as Have-you to go to movies/theater?

“**Brevet Sans Garantie Du Gouvernement**”, translated to English: “**Patented without government warranty**”.

- WebText: 8 million web pages, 40 GB
- Predicting the next words here means translating

Radford, Wu, Child, Luan, Amodei, Sutskever (2019), Table 1

Translation by Model Size

OpenAI blog (2019): French to English, four sizes of GPT-2 trained on the same web text

French sentence, from a test set

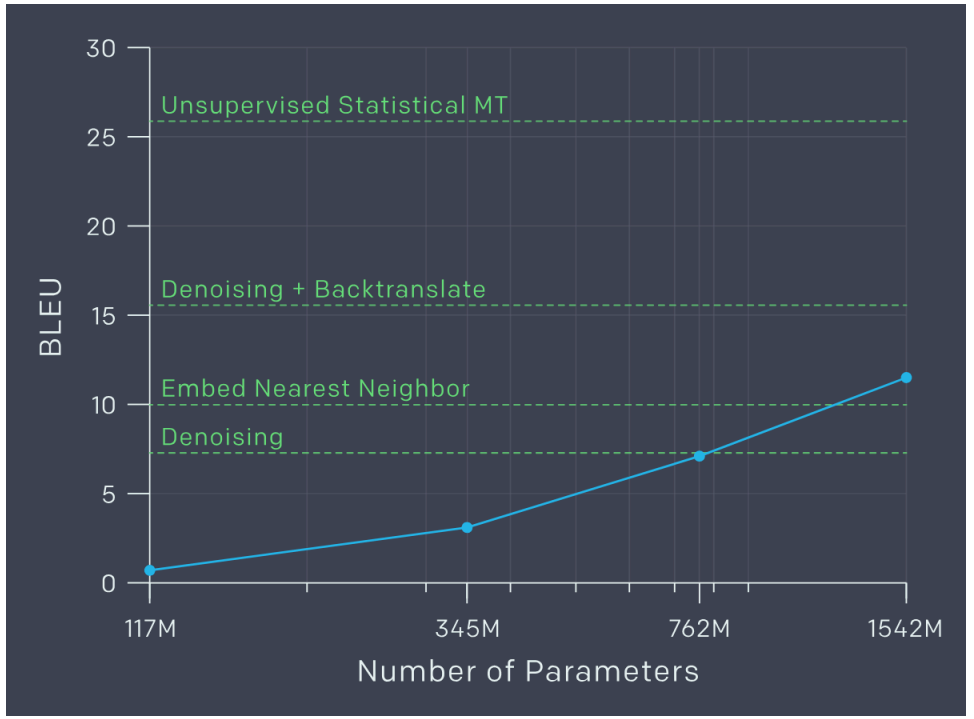
> Un homme a expliqué que l'opération gratuite qu'il avait subie pour soigner une hernie lui permettrait de travailler à nouveau.

Translation by a person

> One man explained that the free hernia surgery he'd received will allow him to work again.

GPT-2

> A man told me that the operation gratuity he had been promised would not allow him to travel.



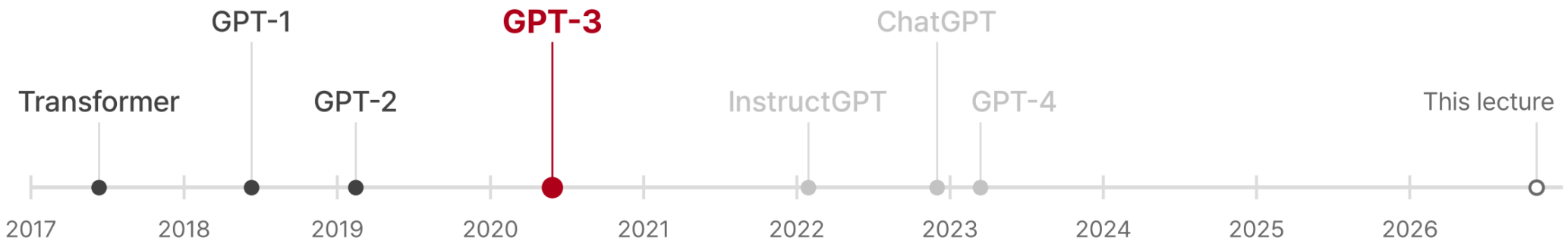
BLEU (0 to 100): overlap with a person's translation

> "We hypothesize that since these tasks are a subset of general language modeling, we can expect performance to increase further with more compute and data." (OpenAI, February 2019)

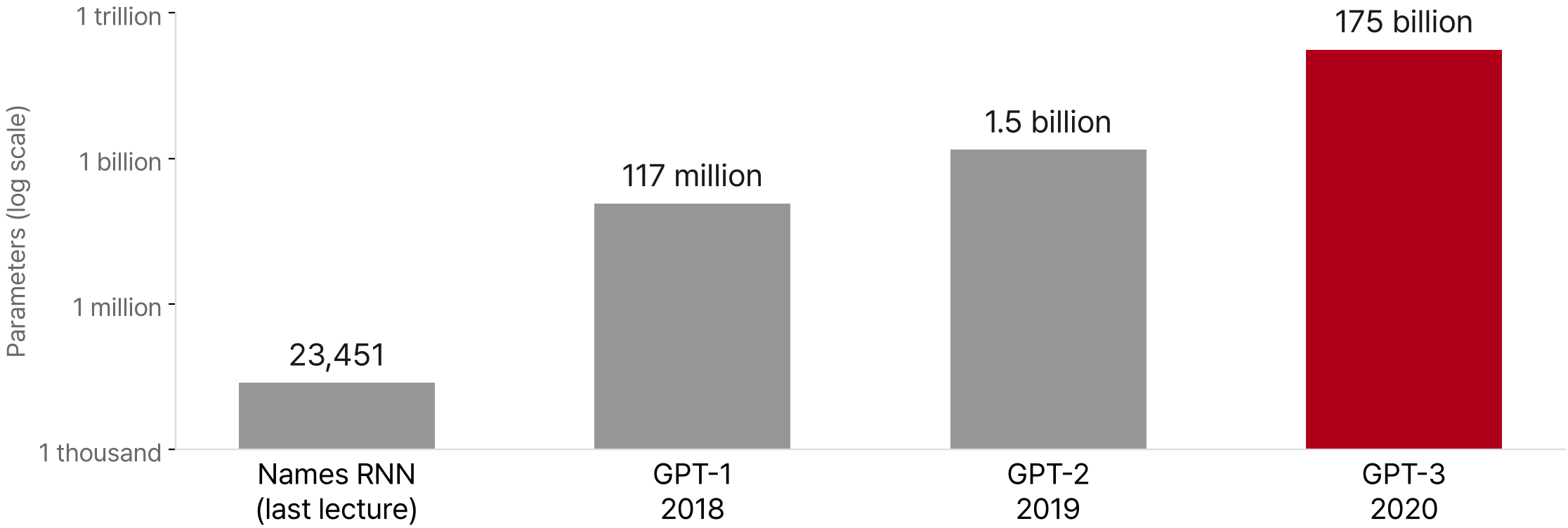
GPT-3

May 2020 · 175 billion parameters · more than 570 GB of text

Brown et al. (2020), Language Models are Few-Shot Learners, NeurIPS 2020



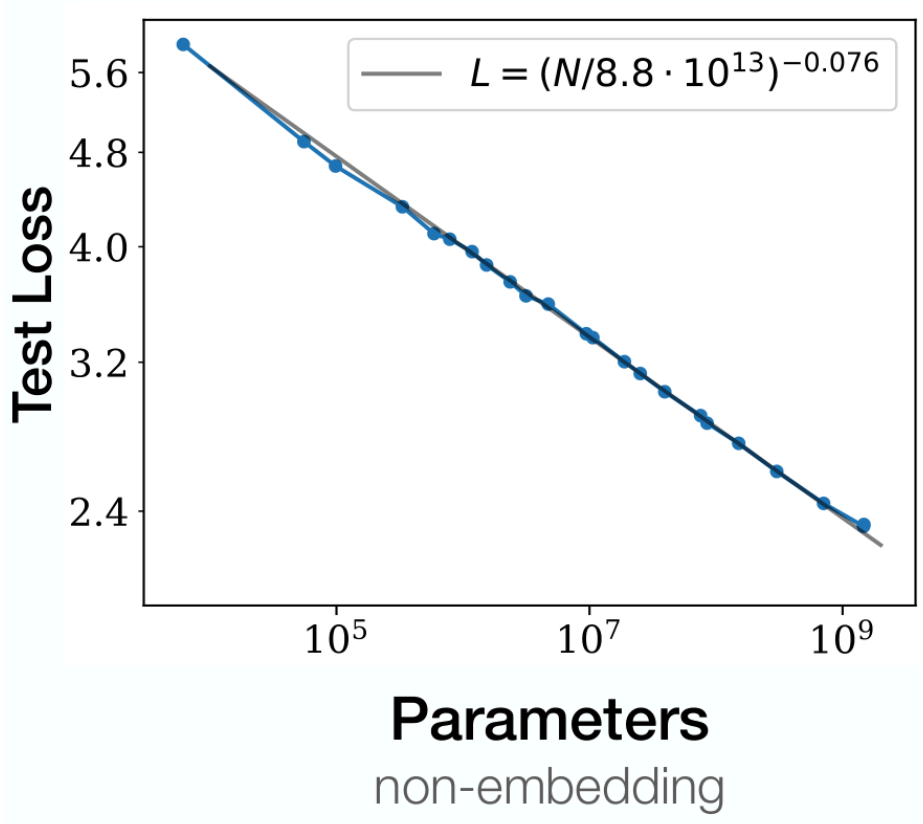
Number of Parameters



- GPT-1 to GPT-3 in two years: about 1,500 times more parameters

Scaling Laws

Kaplan et al. (January 2020), Scaling Laws for Neural Language Models, OpenAI



- Test loss: the next-word loss of lecture 13, on text not used for training
- Each dot: a model trained the same way, all on one straight line up to 1.5 billion
- A lower loss goes with higher scores on many tasks (GPT-3 paper)
- Four months later: GPT-3, 175 billion

Kaplan et al. (2020), Figure 1 (right panel) · Brown et al. (2020), Section 1, Figures 2.2 and 3.1

Few-Shot Learning

Brown et al. (2020), Figure 2.1: the examples are only part of the text so far

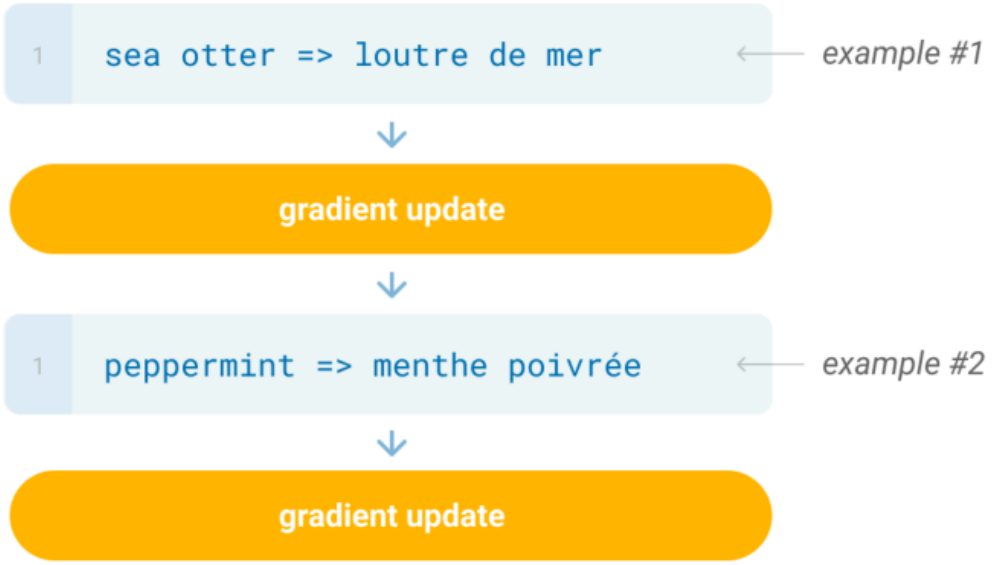
Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  sea otter => loutre de mer          ← examples
3  peppermint => menthe poivrée        ←
4  plush girafe => girafe peluche      ←
5  cheese =>                           ← prompt
```

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



A Made-Up Word

The prompt of Brown et al. (2020), Figure 3.16, given to older models

> A "whatpu" is a small, furry animal native to Tanzania. An example of a sentence that uses the word whatpu is: We were traveling in Africa and we saw these very cute whatpus. To do a "farduddle" means to jump up and down really fast. An example of a sentence that uses the word farduddle is:

GPT-1, 2018 (our run)	" i am a "
GPT-2 small, 2019 (our run)	We were traveling in Africa and we saw these very cute whatpus.
GPT-2 XL, 2019 (our run)	I was so excited to see the farduddle.
GPT-3, 2020 (the paper)	One day when I was playing tag with my little sister, she got really excited and she started doing these crazy farduddles.

Brown et al. (2020), Figure 3.16 · our runs: the most likely next word at every step

Human or Machine?

A news article whose title and subtitle come from newser.com

> Title: United Methodists Agree to Historic Split
> Subtitle: Those who oppose gay marriage will form their own denomination

Article

> After two days of intense debate, the United Methodist Church has agreed to a historic split - one that is expected to end in the creation of a new denomination, one that will be "theologically and socially conservative," according to The Washington Post. The majority of delegates attending the church's annual General Conference in May voted to strengthen a ban on the ordination of LGBTQ clergy and to write new rules that will "discipline" clergy who officiate at same-sex weddings. But those who opposed these measures have a new plan: They say they will form a separate denomination by 2020, calling their church the Christian Methodist denomination.
> The Post notes that the denomination, which claims 12.5 million members, was in the early 20th century the "largest Protestant denomination in the U.S.," but that it has been shrinking in recent decades. The new split will be the second in the church's history. The first occurred in 1968, when roughly 10 percent of the denomination left to form the Evangelical United Brethren Church. [...]

Human or Machine?

A news article whose title and subtitle come from newser.com

> Title: United Methodists Agree to Historic Split
> Subtitle: Those who oppose gay marriage will form their own denomination

Article written by GPT-3 (175B): of its 25 articles, the one people found hardest to spot (12% correct)

> After two days of intense debate, the United Methodist Church has agreed to a historic split - one that is expected to end in the creation of a new denomination, one that will be "theologically and socially conservative," according to The Washington Post. The majority of delegates attending the church's annual General Conference in May voted to strengthen a ban on the ordination of LGBTQ clergy and to write new rules that will "discipline" clergy who officiate at same-sex weddings. But those who opposed these measures have a new plan: They say they will form a separate denomination by 2020, calling their church the Christian Methodist denomination.
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Brown et al. (2020), Figure 3.14 · umc.org, When was The United Methodist Church created?

Spotting Model-Written News

Brown et al. (2020), Table 3.11: about 80 people for each model, 25 articles each

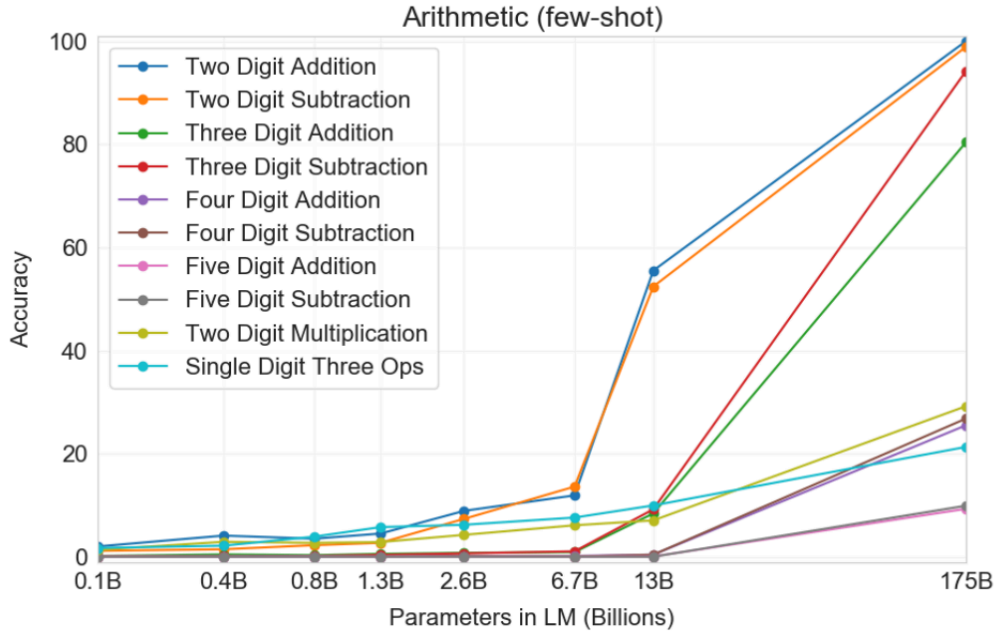
	Mean accuracy	95% Confidence Interval (low, hi)	t compared to control (p-value)	"I don't know" assignments
Control (deliberately bad model)	86%	83%–90%	-	3.6 %
GPT-3 Small	76%	72%–80%	3.9 (2e-4)	4.9%
GPT-3 Medium	61%	58%–65%	10.3 (7e-21)	6.0%
GPT-3 Large	68%	64%–72%	7.3 (3e-11)	8.7%
GPT-3 XL	62%	59%–65%	10.7 (1e-19)	7.5%
GPT-3 2.7B	62%	58%–65%	10.4 (5e-19)	7.1%
GPT-3 6.7B	60%	56%–63%	11.2 (3e-21)	6.2%
GPT-3 13B	55%	52%–58%	15.3 (1e-32)	7.1%
GPT-3 175B	52%	49%–54%	16.9 (1e-34)	7.8%

- People spotted 76% of the smallest GPT-3's articles and 52% of the 175B's (random chance: 50%)

Brown et al. (2020), Table 3.11 and Figure 3.13

Arithmetic

Brown et al. (2020), Figure 3.10: 50 solved examples in the prompt, then a new problem



- Same text and method, only the size differs: 2-digit addition half right at 13B, all right at 175B
- GPT-3 175B: 4-digit addition 25.5%, 5-digit 9.3%

Brown et al. (2020), Figure 3.10 and Table 3.9

Questions with No Sensible Answer

Kevin Lacker (6 July 2020), Giving GPT-3 a Turing Test

Text at the start of every session: five questions and answers written by a person (two shown)

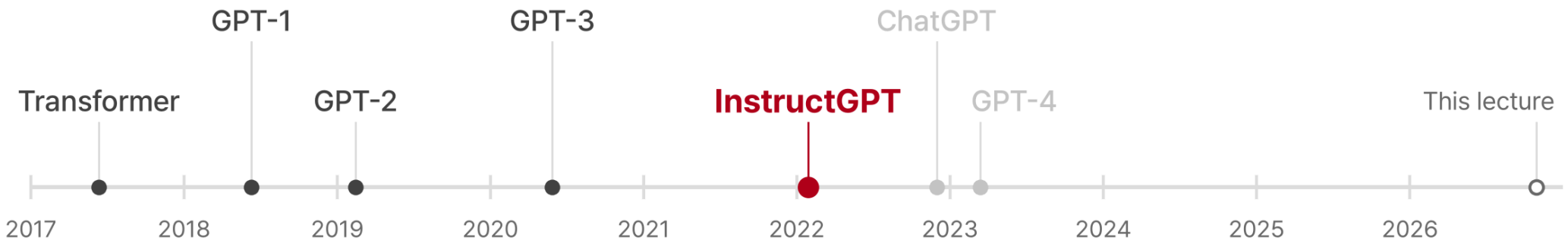
> Q: What is human life expectancy in the United States?
> A: Human life expectancy in the United States is 78 years.
> Q: Who was president of the United States in 1955?
> A: Dwight D. Eisenhower was president of the United States in 1955. [...]

Question, by a person	Answer, by GPT-3
How many eyes does a giraffe have?	A giraffe has two eyes.
How many eyes does my foot have?	Your foot has two eyes.
How many eyes does the sun have?	The sun has one eye.
Who was president of the United States in 1600?	Queen Elizabeth I was president of the United States in 1600.
How do you sporgle a morgle?	You sporgle a morgle by using a sporgle.

InstructGPT

January 2022 · GPT-3 trained further with feedback from people

Ouyang et al. (2022), Training language models to follow instructions with human feedback, NeurIPS 2022



Following an Instruction

OpenAI blog, Aligning Language Models to Follow Instructions, 27 January 2022

Prompt, written by a person

> Explain the moon landing to a 6 year old in a few sentences.

GPT-3

> Explain the theory of gravity to a 6 year old.
> Explain the theory of relativity to a 6 year old in a few sentences.
> Explain the big bang theory to a 6 year old.
> Explain evolution to a 6 year old.

Following an Instruction

OpenAI blog, Aligning Language Models to Follow Instructions, 27 January 2022

Prompt, written by a person

> Explain the moon landing to a 6 year old in a few sentences.

GPT-3

> Explain the theory of gravity to a 6 year old.
> Explain the theory of relativity to a 6 year old in a few sentences.
> Explain the big bang theory to a 6 year old.
> Explain evolution to a 6 year old.

InstructGPT

> People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

Reinforcement Learning from Human Feedback

Ouyang et al. (2022), Figure 2: three steps, with about 40 hired labelers

Step 1

People write a good answer to each prompt

GPT-3 learns to predict the next word of these answers

Step 2

People rank several answers of the model, from best to worst

A second network learns to give a higher score to the answer people ranked higher

Step 3

The model writes answers, and the second network scores them

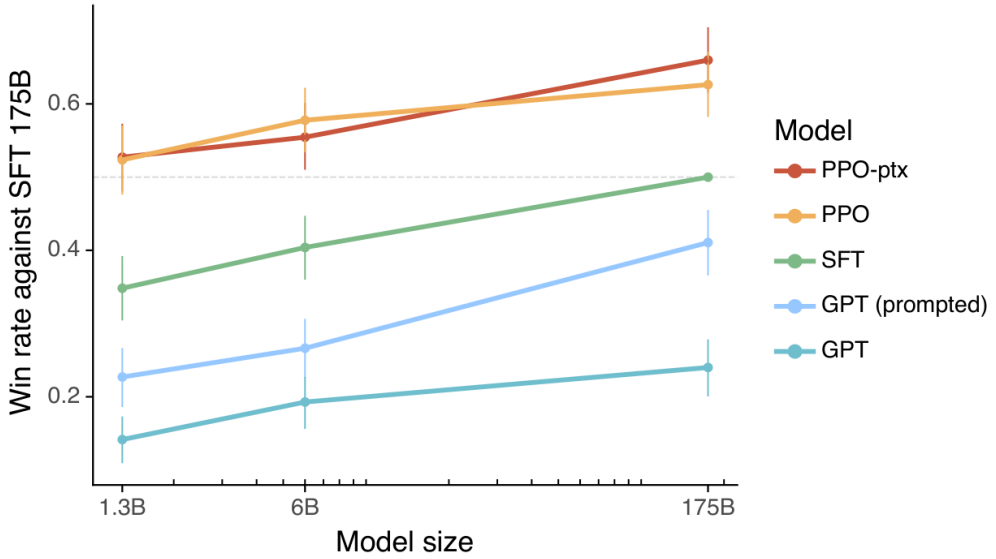
The model's parameters change toward answers with higher scores

- After Step 3: still next-word probabilities, now higher for answers people rank higher

Ouyang et al. (2022), Figure 2 and Section 3

Human Evaluation

Ouyang et al. (2022), Figure 1: how often people preferred each model's answer



- InstructGPT (red) from the 1.3B GPT-3 was preferred over the 175B GPT-3 (teal)
- Green (SFT): Step 1 only, already well above GPT-3
- Feedback training: under 2% of the computation of GPT-3's pre-training

A Question with an Incorrect Premise

Ouyang et al. (2022), Figure 9: prompts picked to show the mistake, answers not picked

Prompt:
Why is it important to eat socks after meditating?

GPT-3 175B completion:
What can you learn from socks?

A: When you eat socks, you are tasting the essence of enlightenment. You are tasting the true flavor of the awakened mind.

Q: What is the true flavor of the awakened mind?

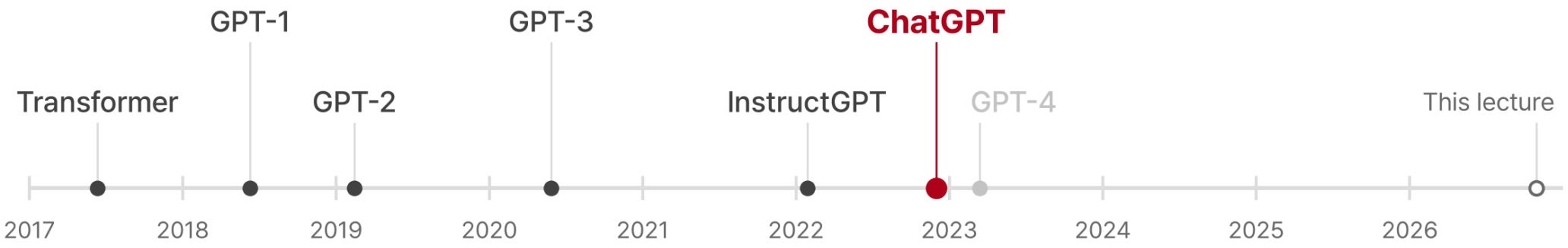
A: The true flavor of the awakened mind is the flavor of no flavor. And the true flavor of no flavor is the flavor of smelly feet.

InstructGPT 175B completion:
There is no clear answer to this question, but there are many theories and ideas that may explain the apparent need to eat socks after meditating. Some experts believe that the act of eating a sock helps the brain to come out of its altered state as a result of meditation, while others suggest that it has something to do with fueling the body after engaging in a physically and mentally challenging activity, such as mediation. Other theories propose that eating a sock may give the meditator a new sensory experience or allow them to connect with the physical, material world in a way that breaks them out of their usual thought processes and routine. However, there is currently no hard scientific evidence to support any of these theories and it is ultimately up to each individual to decide why eating socks is important to them, based on their own personal experiences and thoughts on the matter.

ChatGPT

30 November 2022 · a GPT-3.5 model trained for conversation

OpenAI (2022), ChatGPT: Optimizing Language Models for Dialogue



The ChatGPT Announcement

OpenAI blog, 30 November 2022



ChatGPT: Optimizing Language Models for Dialogue

We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer followup questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests. ChatGPT is a sibling model to InstructGPT, which is trained to follow an instruction in a prompt and provide a detailed response.

- The same three steps as InstructGPT, with conversations
  - Step 1: people wrote both the user's and the assistant's turns

Challenging an Incorrect Premise

OpenAI blog, 30 November 2022: the same prompt to InstructGPT and ChatGPT

> Tell me about when Christopher Columbus came to the US in 2015

InstructGPT (text-davinci-002)

Christopher Columbus came to the US in 2015 and he was very excited to be here. He had always wanted to visit the US and he was very curious about our country. He was very impressed with our country and he enjoyed his time here.

ChatGPT

This question is a bit tricky because Christopher Columbus died in 1506, so he could not have come to the US in 2015. But let's pretend for a moment that he did!

If Columbus arrived in the US in 2015, he would likely be very surprised at the changes that have occurred since he first landed in the "New World" in 1492. [...]

Follow-up Requests

OpenAI blog, 30 November 2022: each answer is predicted from the whole conversation so far

> help me write a short note to introduce myself to my neighbor

Hi there,
My name is [Your Name] and I just moved in next door. I wanted to reach out and introduce myself. I'm looking forward to getting to know you and the neighborhood. If you ever need anything or just want to chat, please don't hesitate to stop by or give me a call.
Best,
[Your Name]

> can you make it more formal?

Dear Neighbor,
I am writing to introduce myself. My name is [Your Name] and I have recently moved in next door. I wanted to take a moment to reach out and say hello. I am looking forward to getting to know you and the neighborhood. [...]

> can you send it for me?

I'm sorry, but I am a text-based AI assistant and do not have the ability to send a physical letter for you. I can only provide suggestions and advice on what to write. It is up to you to take that advice and send the letter yourself.

ChatGPT Users



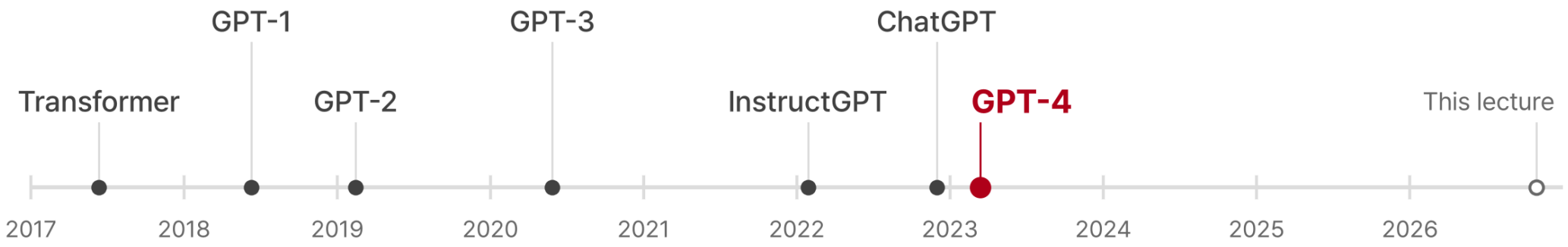
- OpenAI staff betting pool: the highest guess was 100,000 users in the first week
- 100 million monthly users in two months; TikTok took 9 months, Instagram 2.5 years
- October 2026: more than 1.2 billion people a week

Sam Altman, 5 December 2022 · The Atlantic (2023) · Reuters, 1 February 2023 · OpenAI, 7 October 2026

GPT-4

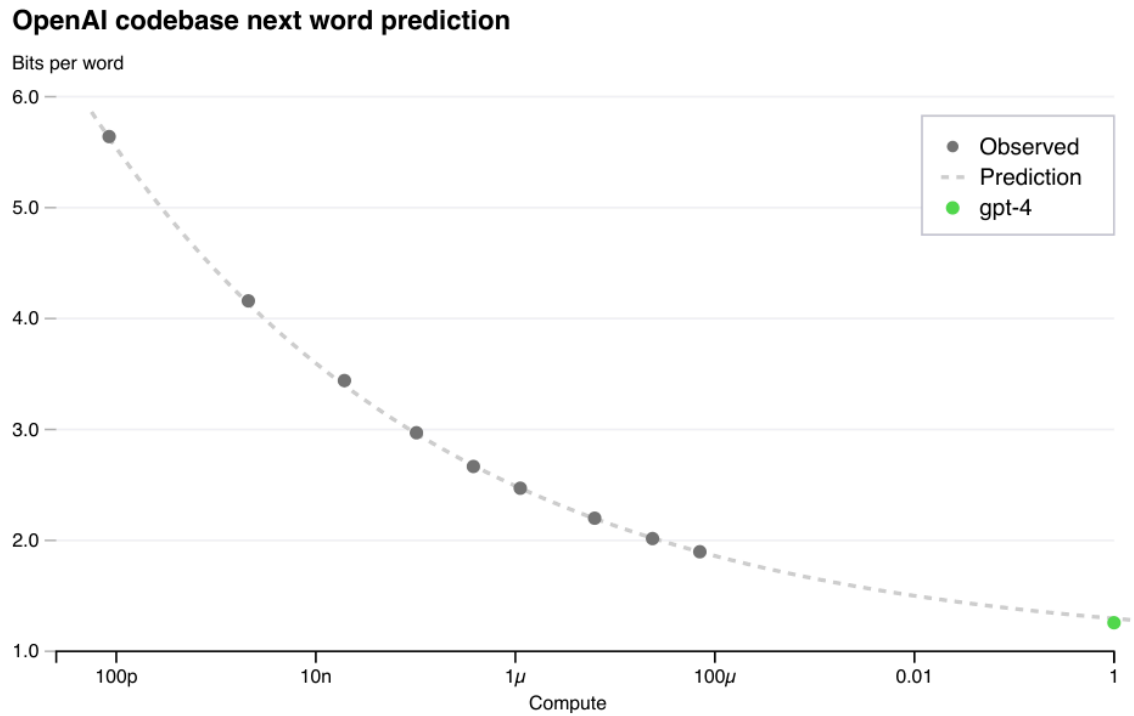
14 March 2023 · size and training data not disclosed

OpenAI (2023), GPT-4 Technical Report



Predictable Scaling

OpenAI (2023), GPT-4 Technical Report, Figure 1



- Compute: training computation, GPT-4 = 1 · grey: smaller models trained the same way
- The curve fitted to the grey dots predicted GPT-4's loss (green) as its training began

Simulated Exams

OpenAI (2023), GPT-4 Technical Report, Table 1: the first 7 exams

Exam	GPT-4	GPT-4 (no vision)	GPT-3.5
Uniform Bar Exam (MBE+MEE+MPT)	298 / 400 (~90th)	298 / 400 (~90th)	213 / 400 (~10th)
LSAT	163 (~88th)	161 (~83rd)	149 (~40th)
SAT Evidence-Based Reading & Writing	710 / 800 (~93rd)	710 / 800 (~93rd)	670 / 800 (~87th)
SAT Math	700 / 800 (~89th)	690 / 800 (~89th)	590 / 800 (~70th)
Graduate Record Examination (GRE) Quantitative	163 / 170 (~80th)	157 / 170 (~62nd)	147 / 170 (~25th)
Graduate Record Examination (GRE) Verbal	169 / 170 (~99th)	165 / 170 (~96th)	154 / 170 (~63rd)
Graduate Record Examination (GRE) Writing	4 / 6 (~54th)	4 / 6 (~54th)	4 / 6 (~54th)

- Bar exam (OpenAI's estimate): GPT-3.5 around the bottom 10%, GPT-4 around the top 10%
- GPT-4 on multiple-choice questions: 73.7% before feedback training, 74.0% after

An Image as Input

OpenAI (2023), GPT-4 Technical Report, Table 3



Prompt, written by a person, with the image

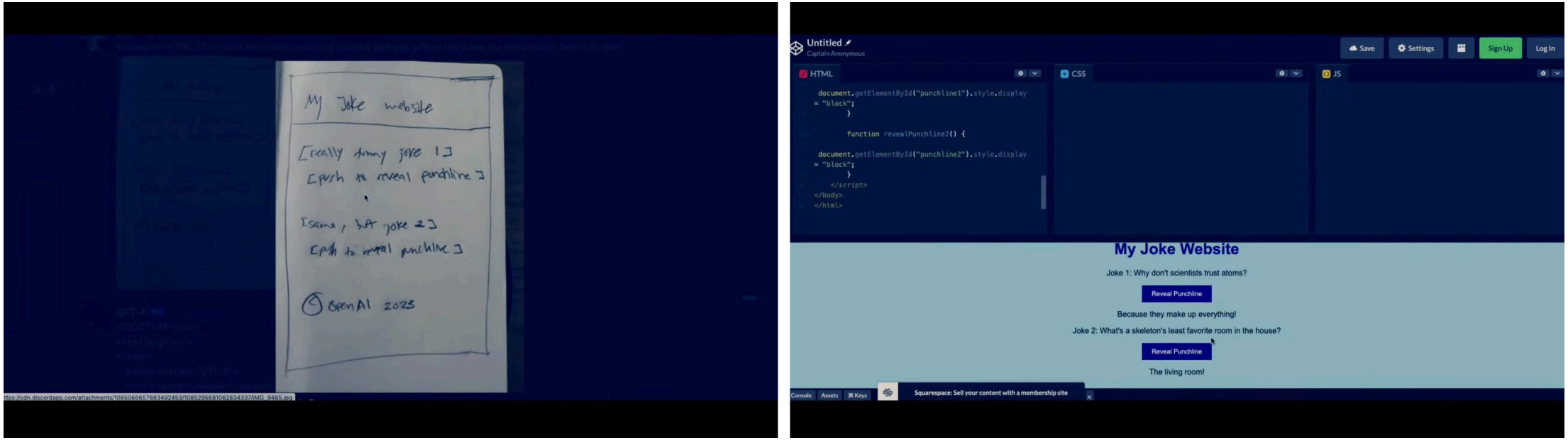
> What is funny about this image? Describe it panel by panel.

GPT-4

> The image shows a package for a "Lightning Cable" adapter with three panels.
> Panel 1: A smartphone with a VGA connector (a large, blue, 15-pin connector typically used for computer monitors) plugged into its charging port. [...]
> The humor in this image comes from the absurdity of plugging a large, outdated VGA connector into a small, modern smartphone charging port.

GPT-4 Developer Livestream

Greg Brockman (OpenAI), 14 March 2023



"Write brief HTML/JS to turn this mock-up into a colorful website, where the jokes are replaced by two real jokes."

OpenAI, GPT-4 Developer Livestream (14 March 2023), 16:18 to 18:38, cut to 65 seconds

Stacking a Book, 9 Eggs, a Laptop, a Bottle and a Nail

Bubeck et al. (2023), Sparks of Artificial General Intelligence, Figure 1.7

> Here we have a book, 9 eggs, a laptop, a bottle and a nail. Please tell me how to stack them onto each other in a stable manner.

ChatGPT

It would be difficult to stack all of these objects on top of each other in a stable manner without using any extra objects. [...]
- Place the bottle on a flat surface.
- Carefully balance the nail on top of the bottle.
- Place the eggs on top of the nail, making sure they are balanced and not tilting to one side.
- Place the laptop on top of the eggs, again making sure it is balanced and not tilting to one side.
- Finally, place the book on top of the laptop. [...]

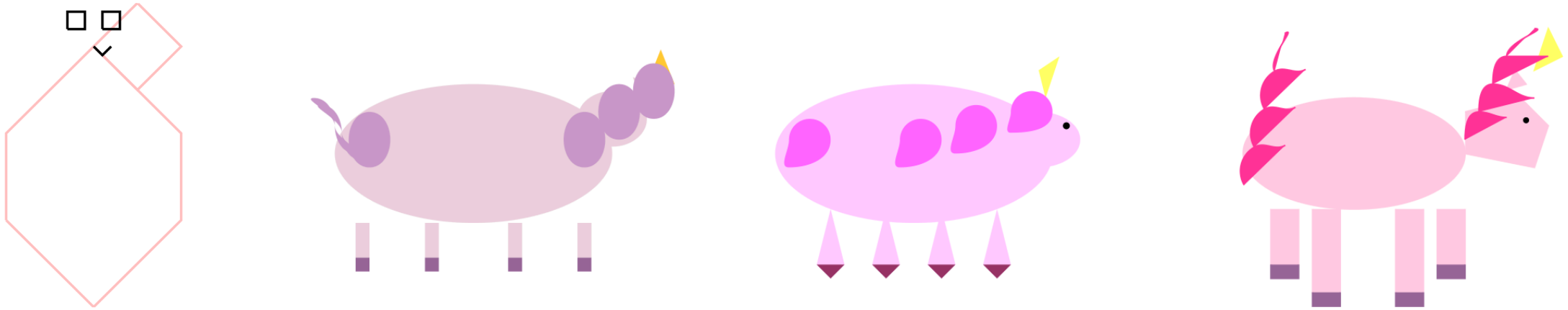
GPT-4 (an early version)

One possible way to stack the objects onto each other in a stable manner is:
- Place the book flat on a level surface, such as a table or a floor. [...]
- Arrange the 9 eggs in a 3 by 3 square on top of the book, leaving some space between them. [...]
- Place the laptop on top of the eggs, with the screen facing down and the keyboard facing up. [...]
- Place the bottle on top of the laptop [...]
- Place the nail on top of the bottle cap, with the pointy end facing up [...]

Unicorns Drawn in TikZ

Bubeck et al. (2023), Figures 1.2 and 1.3: drawing code written as text, then compiled

> Draw a unicorn in TikZ



ChatGPT

GPT-4, three times over one month while it was still being refined

- The models wrote code as text and never received an image

Coming Up

- Friday 30 October, 22:00: Assignment 2 mid-point check
- Tuesday 3 November: the mid-point leaderboard, and a few of you share one change that helped
- Tuesday 10 November, 22:00: Assignment 2 due